

Shaden Shaar

NLP RESEARCHER · PHD CANDIDATE AT CORNELL UNIVERSITY

Ithaca, NY, USA

+1 (646) 286-3453 | ✉ sshaar31[at]gmail[dot]com | 🏠 sshaar.github.io | 📷 sshaar | 🎓 Google Scholar

About Me

A PhD Candidate in Computer Science at Cornell University, advised by Prof. Claire Cardie, working on **long-form generation in multi-modal settings** — specifically long-form question-answering and summarization over long videos. In parallel, I collaborate with NewYork-Presbyterian Hospital on **clinical NLP** for heart failure and heart transplant patients. Previously, at the Qatar Computing Research Institute, I worked with Prof. Preslav Nakov on automated fact-checking, propaganda detection, and COVID-19 misinformation.

Research Interests

- **Multi-modal long-form generation.** Long-form question-answering and summarization over videos and other multi-modal narratives.
- **Clinical NLP.** Using LLMs to surface clinical decision patterns from unstructured medical narratives, with a focus on heart failure and heart transplant care.
- **Automated fact-checking** (*prior*). Detecting, verifying, and justifying claim veracity across multi-modal, variable-length inputs.
- **Propaganda detection** (*prior*). Identifying propaganda and persuasion techniques across multi-modal inputs in news articles, memes, and other media.

Highlights

- **h-index 24** with ~2,858 citations on Google Scholar.
- **Best Demo Award (Honorable Mention)** at ACL 2020 for *Prta*, a system for analyzing propaganda techniques in news.
- **University Fellowship** at Cornell University (2021) for exceptional preparation and promise.
- **University Honors** at Carnegie Mellon University (2019); **50% Academic Merit Scholarship** at CMU (2015); five-time Dean's List awardee.
- Co-organized multiple shared tasks at **CLEF-CheckThat!** (2020, 2021) and **SemEval** (2021) on automated fact-checking and propaganda detection.

Work Experience

Zillow Group

APPLIED SCIENTIST INTERN

Remote, USA

May 2025 – Aug. 2025

- Built conversational assistive AI agents for Zillow's real-estate platform using reinforcement-learning methods.
- Designed reward-modeling and evaluation pipelines for grounded multi-turn property-search dialogue.

ScaleAI, Inc.

MACHINE LEARNING RESEARCH ENGINEER INTERN

New York City, NY, USA

Jan. 2025 – May 2025

- Built an Arabic-language AI assistant for the Qatari judicial department to aid judges reviewing active cases.
- Implemented dense and sparse retrieval over Arabic cassation and supreme-court rulings for relevant-precedent lookup.
- Designed Arabic case-summarization and event-extraction pipelines feeding the judge-facing platform.

Apple

AI/ML RESEARCH INTERN

Seattle, WA, USA

May 2022 – Aug. 2022

- Worked with Dr. Alex Churchill on zero-shot multi-turn conversation data generation.
- Built a model that generates conditioned multi-turn dialogue datasets for downstream Siri training.

Qatar Computing Research Institute

RESEARCH ASSISTANT

Doha, Qatar

Jul. 2019 – Aug. 2021

- Worked with Dr. Preslav Nakov and Prof. Giovanni Da San Martino on fact-checking and propaganda detection.
- Introduced the task of detecting previously fact-checked claims; methods adopted by leading fact-checking organizations.
- Built fact-checking and analysis pipelines for COVID-19 misinformation across social platforms.
- Co-organized shared tasks at CLEF-CheckThat! (2020, 2021) and SemEval-2021 Task 6 on persuasion in memes.
- Built and shipped *Prta*, a public propaganda-analysis demo (ACL 2020 Best Demo, Honorable Mention).

JOURNALS

1. Frye, J., **Shaar, S.**, Cardie, C., DeFilippis, E., Estrin, D., Sayer, G., Uriel, N., et al. "Thematic Analysis of Accepted Exception Requests for Heart Transplant Candidates Using a Large Language Model." *The Journal of Heart and Lung Transplantation* 45(5):460, 2026. (**JHLT'2026**).
2. **Shaar, S.**, Chen, W., Chatterjee, M., Wang, B., Zhao, W., and Cardie, C. "Are Triggers Needed for Document-Level Event Extraction?" *Transactions of the Association for Computational Linguistics* (**TACL'2025**).

CONFERENCES

1. **Shaar, S.**, Thymes, B., Chaixanien, S., Cardie, C., and Hariharan, B. "MovieRecapsQA: A Multimodal Open-Ended Video Question-Answering Benchmark." *Conference on Computer Vision and Pattern Recognition* (**CVPR'2026**).
2. Chaixanien, S., Choi, E., **Shaar, S.**, and Cardie, C. "Pungene at DialAM-2024: Identification of Propositional and Illocutionary Relations." *Workshop on Argument Mining at ACL 2024* (**ArgumentMining@ACL'2024**).
3. Aydin, A. N., **Shaar, S.**, and Cardie, C. "Edward Said at Touché: Human Value Detection Using Transformers and Upsampling." *Working Notes of the Conference and Labs of the Evaluation Forum* (**CLEF'2024**).
4. Nakov, P., Barrón-Cedeño, A., Da San Martino, G., Alam, F., Struß, J. M., Mandl, T., Míguez, R., Caselli, T., Kutlu, M., Zaghouani, W., et al. "The CLEF-2022 CheckThat! Lab on Fighting the COVID-19 Infodemic and Fake News Detection." *European Conference on Information Retrieval* (**ECIR'2022**).
5. Nakov, P., Barrón-Cedeño, A., Da San Martino, G., Alam, F., Struß, J. M., Mandl, T., Míguez, R., Caselli, T., Kutlu, M., Zaghouani, W., et al. "Overview of the CLEF-2022 CheckThat! Lab on Fighting the COVID-19 Infodemic and Fake News Detection." *Cross-Language Evaluation Forum for European Languages* (**CLEF'2022**).
6. **Shaar, S.**, Georgiev, N., Alam, F., Da San Martino, G., Mohamed, A., and Nakov, P. "Assisting the Human Fact-Checkers: Detecting All Previously Fact-Checked Claims in a Document." *Findings of the Association for Computational Linguistics: EMNLP 2022* (**EMNLP'2022**).
7. **Shaar, S.**, Alam, F., Da San Martino, G., and Nakov, P. "The Role of Context in Detecting Previously Fact-Checked Claims." *Findings of the Association for Computational Linguistics: NAACL 2022* (**NAACL'2022**).
8. Nakov, P., Alam, F., **Shaar, S.**, Da San Martino, G., and Zhang, Y. "A Second Pandemic? Analysis of Fake News About COVID-19 Vaccines in Qatar." *Findings of the Association for Computational Linguistics: NAACL 2022* (**NAACL'2022**).
9. Hassan, S., **Shaar, S.**, and Darwish, K. "Cross-Lingual Emotion Detection." *Proceedings of the Thirteenth Language Resources and Evaluation Conference* (**LREC'2022**).
10. Alam, F., Cresci, S., Chakraborty, T., Silvestri, F., Dimitrov, D., Da San Martino, G., **Shaar, S.**, Firooz, H., and Nakov, P. "A Survey on Multimodal Disinformation Detection." *International Conference on Computational Linguistics* (**COLING'2022**).
11. Dimitrov, D., Bin Ali, B., **Shaar, S.**, Alam, F., Silvestri, F., Firooz, H., Nakov, P., and Da San Martino, G. "Detecting Propaganda Techniques in Memes." *Association for Computational Linguistics* (**ACL-IJCNLP'2021**).
12. Dimitrov, D., Bin Ali, B., **Shaar, S.**, Alam, F., Silvestri, F., Firooz, H., Nakov, P., and Da San Martino, G. "SemEval-2021 Task 6: Detection of Persuasion Techniques in Texts and Images." *ACL-IJCNLP SemEval Workshop* (**SemEval@ACL-IJCNLP'2021**).
13. **Shaar, S.**, Alam, F., Da San Martino, G., Nikolov, A., Zaghouani, W., Nakov, P., and Feldman, A. "Findings of the NLP4IF-2021 Shared Tasks on Fighting the COVID-19 Infodemic and Censorship Detection." *ACL NLP4IF Workshop* (**NLP4IF@ACL'2021**).
14. Nakov, P., Da San Martino, G., Elsayed, T., Barrón-Cedeño, A., Míguez, R., **Shaar, S.**, Alam, F., Haouari, F., Hasanain, M., Babulkov, N., Nikolov, A., Shahi, G., Struß, J., and Mandl, T. "The CLEF-2021 CheckThat! Lab on Detecting Check-Worthy Claims, Previously Fact-Checked Claims, and Fake News." *European Conference on Information Retrieval* (**ECIR'2021**).
15. Skuczynska, B., **Shaar, S.**, Spenader, J., and Nakov, P. "BeaSku at CheckThat! 2021: Fine-Tuning Sentence BERT with Triplet Loss and Limited Data." *CLEF Conference and Labs of the Evaluation Forum* 639–647, (**CLEF'2021**).
16. Alam, F., **Shaar, S.**, Dalvi, F., Sajjad, H., Nikolov, A., Mubarak, H., Da San Martino, G., Abdelali, A., Durrani, N., Darwish, K., Al-Homaid, A., Zaghouani, W., Caselli, T., Danoe, G., Stolk, F., Bruntink, F., and Nakov, P. "Fighting the COVID-19 Infodemic: Modeling the Perspective of Journalists, Fact-Checkers, Social Media Platforms, Policy Makers, and the Society." *Findings of EMNLP* (**EMNLP'2021**).
17. Alam, F., Dalvi, F., **Shaar, S.**, Durrani, N., Mubarak, H., Nikolov, A., Da San Martino, G., Abdelali, A., Sajjad, H., Darwish, K., and Nakov, P. "Fighting the COVID-19 Infodemic in Social Media: A Holistic Perspective and a Call to Arms." *International AAAI Conference on Web and Social Media* (**ICWSM'2021**).
18. Nakov, P., Alam, F., **Shaar, S.**, Da San Martino, G., and Zhang, Y. "COVID-19 in Bulgarian Social Media: Factuality, Harmfulness, Propaganda, and Framing." *International Conference on Recent Advances in Natural Language Processing* (**RANLP'2021**).

19. Nakov, P., Corney, D., Hasanain, M., Alam, F., Elsayed, T., Barrón-Cedeño, A., Papotti, P., **Shaar, S.**, and Da San Martino, G. "Automated Fact-Checking for Assisting Human Fact-Checkers." *International Joint Conferences on Artificial Intelligence (IJCAI'2021)*.
20. **Shaar, S.**, Da San Martino, G., Babulkov, N., and Nakov, P. "That Is a Known Lie: Detecting Previously Fact-Checked Claims." *Association for Computational Linguistics (ACL'2020)*.
21. Da San Martino, G., **Shaar, S.**, Zhang, Y., Yu, S., Barrón-Cedeño, A., and Nakov, P. "Prta: A System to Support the Analysis of Propaganda Techniques in the News." *Association for Computational Linguistics (ACL'2020)*. **Best Demo Award (Honorable Mention)**.
22. **Shaar, S.**, Nikolov, A., Babulkov, N., Alam, F., Barrón-Cedeño, A., Elsayed, T., Hasanain, M., Suwaileh, R., Haouari, F., Da San Martino, G., and Nakov, P. "Overview of CheckThat! 2020 English: Automatic Identification and Verification of Claims in Social Media." *CLEF Working Notes* 2696, **(CLEF'2020)**.
23. Barrón-Cedeño, A., Elsayed, T., Nakov, P., Da San Martino, G., Hasanain, M., Suwaileh, R., Haouari, F., Babulkov, N., Hamdan, B., Nikolov, A., **Shaar, S.**, and Ali, Z. S. "Overview of CheckThat 2020: Automatic Identification and Verification of Claims in Social Media." **(CLEF'2020)**, 11:215–236.
24. **Shaar, S.**, Razak, S., Dalvi, F., and Moosavi, S. "Group Identification in Crowded Environments Using Proximity Sensing." *IEEE Conference on Local Computer Networks (LCN'2018)*.
25. Hassan, S., **Shaar, S.**, Raj, B., and Razak, S. "Interactive Evaluation of Classifiers Under Limited Resources." *IEEE International Conference on Machine Learning and Applications (ICMLA'2018)*.

Education

Cornell University

PH.D. IN COMPUTER SCIENCE

Minor: Applied Mathematics.

Ithaca, NY, USA

Aug. 2021 – Present

Cornell University

M.S. IN COMPUTER SCIENCE

Ithaca, NY, USA

Aug. 2021 – May 2024

Carnegie Mellon University

B.S. IN COMPUTER SCIENCE

Minor in Mathematics; University Honors

Doha, Qatar

Aug. 2015 – May 2019

Honors & Awards

- | | | |
|------|---|------------------------|
| 2021 | University Fellowship , Cornell University, awarded for exceptional preparation and promise | <i>Ithaca, NY, USA</i> |
| 2020 | Best Demo Award (Honorable Mention) , ACL 2020, for <i>Prta: A System to Support the Analysis of Propaganda Techniques</i> | <i>Online</i> |
| 2019 | University Honors , Carnegie Mellon University, for outstanding cumulative GPA | <i>Doha, Qatar</i> |
| 2019 | Dean's List (5 awards) , Carnegie Mellon University, F15, S16, F16, F17 | <i>Doha, Qatar</i> |
| 2015 | 50% Academic Merit Scholarship , Carnegie Mellon University, awarded to a select group of incoming students | <i>Doha, Qatar</i> |

Academic Services

- | | | |
|------|---|---------------------------|
| 2021 | Co-organizer , SemEval–2021 Task 6: Detection of Persuasion Techniques in Texts and Images | <i>Online</i> |
| 2021 | Co-organizer , CLEF–2021 CheckThat! Lab on detecting check-worthy and previously fact-checked claims | <i>Bucharest, Romania</i> |
| 2020 | Co-organizer , CLEF–2020 CheckThat! Lab on automatic identification and verification of claims | <i>Online</i> |
| 2020 | Reviewer , ACL, EMNLP, NAACL, EACL, COLING, AAAI, CIKM (Program Committee) | |

Teaching Experience

Cornell University

Ithaca, NY, USA

TEACHING ASSISTANT

2021 – 2022

- CS 4740 – Natural Language Processing (Fall 2022, Lecturer: Prof. Lillian Lee).
- CS 5780 – Introduction to Machine Learning (Spring 2022).

Carnegie Mellon University

Doha, Qatar

TEACHING ASSISTANT

2016 – 2019

- 11-785 – Introduction to Deep Learning (Fall 2018, Spring 2019, Lecturer: Prof. Bhiksha Raj).
- 15-251 – Great Theoretical Ideas in Computer Science (Spring 2018, Spring 2019).
- 15-213 – Introduction to Computer Systems (Fall 2017).
- 15-112 – Fundamentals of Programming (Fall 2016, Fall 2017).
- 21-127 – Concepts of Mathematics (Spring 2016).